

Question ID: 25755def

Question

While researching a topic, a student has taken the following notes:

- Generally, an object will heat up when twisted.
- The twisting of an object is known as torsion.
- A 2019 study led by Zunfeng Liu and Ray Baughman tested the torsional heating of various fibers.
- When a 3-millimeter-thick sample of thermoplastic polyurethane (TPU) fiber was twisted, its average surface temperature increased by 6°C.
- When a 4-millimeter-thick sample of styrene-ethylene-butylene-styrene (SEBS) rubber fiber was twisted, its average surface temperature increased by 3.5°C.

The student wants to contrast the two samples. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

- A. When the fibers were twisted as part of the 2019 study, the surface temperature of both samples increased.
- B. In 2019, researchers studied the effect of torsional heating on various fibers, including samples of SEBS rubber and TPU.
- C. Twisting an object will generally cause its temperature to increase, a process known as torsional heating.
- D. The SEBS rubber sample used in the 2019 study was thicker than the TPU sample.